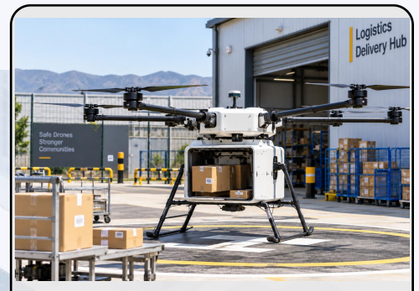
(a) UAV Logistics Operations under Disruptions



Cargo handling

E Cross-dock

U02

C → E (normal route)

D Clinic

Charging Coordination

C → D delayed (charging)

A → C closed (disruption)

U01

A → B → C (reroute)

C → F delayed (charging)

A Logistics Depot

B Delivery Station

F Parcel Lockers

1 **Disruption** A → C closed

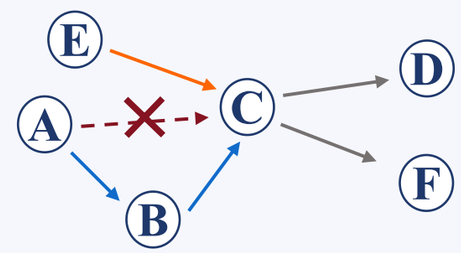
2 **Reroute** U01 via B

3 **Time & Energy Shift** $\triangle t \uparrow, E_{arr} \downarrow$

4 **Charging Coordination** U01 ↔ U02

5 **Downstream impact** C→D,C→F delayed

(b) Directed Network



(c) Fixed Mission Chains

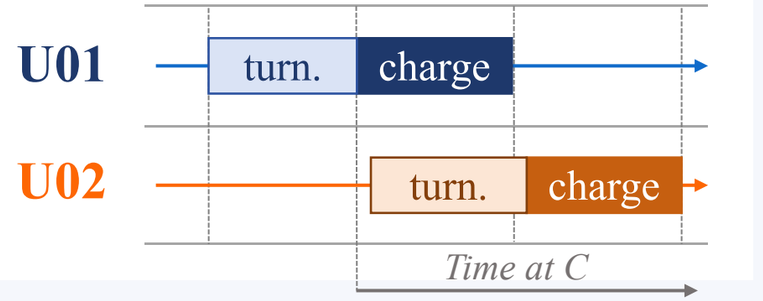
U01 *suffix hold* h_m

m_1: A→C	m_2: C→D	m_3: D→F

U02

m_4: E→C	m_5: C→F

(d) Shared Charging Position C



(e) ARCO-Bench Evaluation and Scope Diagnosis

1 Benchmark Tasks

- Original networks
- Paired closures
- Core-72
- Disrupted record & priorities

2 Candidate Evaluation

Candidate Repair Model / Solver → **Independent Verifier** → **Verified Record** Feasibility Objective L1-L6

Independent Verifier

Schema & Decisions *(h,p,d,c)* → **Event Reconstruction** *positioning · service turnaround · charging* → **Timing** *release | precedence*; **Battery** *SOC | reserve*; **Resources** *corridor | pad | charger*

3 Reference & Scope

MILP Reference Separate solve

Recovery Target T_j Service / quality

Scope Diagnosis (S,k) Chains · Changes · Values